

ERDŐS PROBLEM 928 AS A COROLLARY OF OPENAI'S JOINT DICKMAN LAW

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ABSTRACT. We identify Erdős Problem 928 as a consequence of OpenAI's joint Dickman law. An elementary counting estimate shows that the different inequality and threshold conventions do not change the natural density.

Let $P(n)$ be the largest prime divisor of the integer $n \geq 2$, and let ρ denote the Dickman function. For fixed $\alpha, \beta \in (0, 1)$ and real $X \geq 2$, put

$$\begin{aligned} A_{\alpha, \beta}(X) &= \#\{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < (n+1)^\beta\}, \\ B_{\alpha, \beta}(X) &= \#\{n \in \mathbb{N} : 2 \leq n \leq X, P(n) \leq n^\alpha, P(n+1) \leq n^\beta\}. \end{aligned}$$

Problem 928 in Bloom's collection [1] asks whether $A_{\alpha, \beta}(X)/X$ has a limit, and records the predicted value $\rho(1/\alpha)\rho(1/\beta)$.

OpenAI's joint Dickman law [2, Theorem 1.1, p. 2] (result family 012 of its mathematics release) gives

$$(1) \quad \lim_{X \rightarrow \infty} \frac{B_{\alpha, \beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

Corollary 1. *The joint Dickman law (1) implies that, for every $\alpha, \beta \in (0, 1)$,*

$$\lim_{X \rightarrow \infty} \frac{A_{\alpha, \beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

Proof. An integer counted by $B_{\alpha, \beta}(X)$ but not by $A_{\alpha, \beta}(X)$ must satisfy $P(n) = n^\alpha$, since $n^\beta < (n+1)^\beta$. On this exceptional set, the map $n \mapsto P(n)$ is injective and takes positive integer values at most X^α . There are therefore at most $\lfloor X^\alpha \rfloor$ such integers.

Conversely, an integer counted by $A_{\alpha, \beta}(X)$ but not by $B_{\alpha, \beta}(X)$ must satisfy

$$n^\beta < P(n+1) < (n+1)^\beta.$$

The open intervals $(n^\beta, (n+1)^\beta)$ are pairwise disjoint as n varies. Thus $n \mapsto P(n+1)$ is injective on this exceptional set, and its values are positive integers below $(X+1)^\beta$. Its cardinality is at most $\lfloor (X+1)^\beta \rfloor$. The two bounds give, for every $X \geq 2$,

$$(2) \quad |A_{\alpha, \beta}(X) - B_{\alpha, \beta}(X)| \leq \lfloor X^\alpha \rfloor + \lfloor (X+1)^\beta \rfloor.$$

Dividing by X , the right-hand side tends to zero because $\alpha, \beta < 1$; the conclusion follows from (1). $\square$

The higher-order independence conjecture for three or more consecutive integers mentioned in [1] is outside the scope of this corollary.

REFERENCES

- [1] T. F. Bloom, *Erdős Problem #928*, Erdős Problems, <https://www.erdosproblems.com/928>, accessed October 9, 2026.
- [2] OpenAI, *The joint Dickman law for consecutive integers*, preprint, September 24, 2026, result family 012; Theorem 1.1, p. 2. OpenAI Math Release, repository version `fd4aeeb2ee4f`.